

Aaditya Vikram Saravana Bhavan

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Education

Carnegie Mellon University

Pittsburgh, PA

Master of Computational Data Science (MCDS), GPA: 4.08/4.00

Dec 2026

Relevant Courses: Introduction to Deep Learning (PhD), Introduction to Machine Learning (MS), Foundations of Computational Data Science (MS), Cloud Computing (MS)

PES University

Bangalore, India

B.Tech in Computer Science and Engineering, GPA: 9.42/10.00

Relevant Courses: Data Structures and Algorithms, Computer Networks, Data Analytics, Big Data

Skills

Programming and Problem Solving: C/C++ (advanced), Go (advanced), Python (proficient), Java (moderate), SQL; **Knight at LeetCode (Top 2.75% globally)**

Data Science and Machine Learning: Pattern Recognition, Data Mining, Big Data Analytics, PyTorch, pandas, numpy, scikit-learn, TensorFlow, Keras, HuggingFace, NLTK, Information Retrieval

Software Development and Database: HTML, CSS, React.js, Express.js, Node.js, MongoDB Atlas, PostgreSQL, Unix/Linux Environments

DevOps and Tools: Git, GitHub, Jenkins, Docker, Kubernetes, Grafana, Jira, Confluence

Experience

Cisco

Chennai, India

Software Development Engineer at Catalyst Center

Sep 2024 – Jul 2025

- Led development of a Go based telemetry middleware, validating pipelines via AWS Athena for 10K+ customers; developed Webex bots (Node.js) for real-time alerts, reducing MTTR by 40%
- Improved platform reliability by 35% by fixing 50+ backend/UI issues and standardizing API error handling across 20+ microservices

Zscaler

Bangalore, India

Software Development Engineer Intern

Feb 2024 – Aug 2024

- Reduced data size by 40% and transfer latency by 50% using Protobufs; optimized PostgreSQL/PL-SQL workloads
- Secured RabbitMQ communication, automated server workflows with Python, and achieved 95% unit test coverage

Undergraduate Teaching Assistant

Bangalore, India

UE20CS351 — Cloud Computing

Aug 2023 – Dec 2023

- Led Docker and ETCD labs and designed coursework for 600+ students, improving learning outcomes

Zscaler

Bangalore, India

Software Development Engineer Intern

Jun 2023 – Aug 2023

- Worked across system level C code, designing efficient serialization pipelines, reducing code duplication(X-Macros)

Academic Projects

MyTorch

Pittsburgh, PA

Carnegie Mellon University, Intro to Deep Learning

Aug 2025 – Dec 2025

- Built a PyTorch like deep learning library from scratch, implementing forward/backward propagation, autograd style training, and models including MLPs, CNNs, RNNs (GRU/LSTM), and Transformers

GAN-Based Multispectral Image Dehazing

Bangalore, India

PES University

Jan 2023 – Apr 2024

- Developed a GAN-based model in TensorFlow/Python for multispectral image dehazing, outperforming existing methods on the SHIA dataset
- Awarded Best Paper (ISMSI 2024) for advancing state of the art image dehazing techniques

Yet Another Kafka

Bangalore, India

PES University

Nov 2022 – Jan 2023

- Built a Kafka-like distributed messaging system using socket programming with producer-consumer architecture
- Implemented multi-broker replication with Zookeeper recovery, enabling fault tolerance and client subscriptions

Plan Your Miles - Travel Website

Bangalore, India

PES University

Sep 2021 – Jan 2022

- Led a 4 member team to build a MERN stack web application for personalized vacation planning using user preference capture and emotion-based analysis
- Improved recommendation accuracy by 25% and user engagement by 30% in evaluations

Publications

- **Enhancing Multispectral Vision: A GAN-Based Dehazing Framework**, ISMSI 2024
- **A Pugnacious Comparative Study of Data Analysis Techniques for Wine Quality**, I2CT 2024
- **Android Malware Detection: A Comprehensive Review**, Research Advances in Network Technologies 2024